

Topgyal Gurung

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Full Stack Software Engineer with a bachelor's in computer science and strong backend and database expertise. Experienced in designing scalable systems, deploying full stack applications, and developing secure APIs. Skilled in JavaScript, TypeScript, Python, SQL, and modern frameworks, with hands-on experience in data engineering, and security-focused development. Actively developing cloud and AI skills including AWS and LLM integrations.

Technical Skills

PROGRAMMING & LANGUAGES: JavaScript, TypeScript, Python, SQL, HTML, CSS, Shell Scripts

FRAMEWORKS & LIBRARIES: React, Redux, Next.js, Node.js, Express, Docker, Kubernetes

DATABASES & CLOUD: PostgreSQL, MongoDB, Redis, GCP BigQuery, AWS

APIs & TOOLS: REST, GraphQL, Google Maps, Cloudinary, Stripe, Git, GitHub Actions (CI/CD), TDD, Vim

AI & EMERGING TECH: Vercel AI SDK, LangChain, OpenAI APIs, LLM application development

CERTIFICATIONS: Python for Everybody, AWS Academy Cloud Foundations, CodePath Cybersecurity, Big Data Analytics with Hadoop & Spark

Project Highlights

Financial Invoice Dashboard | 07/2025

TypeScript, Next.js, React, Auth.js, Tailwind CSS, Postgres.js

- Built a secure full-stack Next.js dashboard with Auth.js authentication, protected routes, streaming Suspense, and loading skeletons, improving user experience by reducing page-blocking delays.
- Optimized data management with search, pagination, caching, and form validation, enabling efficient invoice creation, editing, and deletion while reducing latency.
- Architected backend using Neon Serverless Postgres + Postgres.js, colocating database with app, fetching only required data via SQL, and leveraging React Server Components for faster load times.

Myspace social media clone | 06/2025

React, Next.js, Auth.js, Prisma ORM(PostgreSQL)

- Built a full-stack social media platform applying advanced Next.js 14 concepts from Fireship's course, including multiple rendering modes, server components and data fetching strategies.
- Implemented features such as follow/unfollow, Google OAuth authentication (Auth.js), profile management and real time feed updates, optimized rendering, and caching for a smooth, scalable user experience.

CunyFirst2.0 Course Scheduler | 06/2019

React, Node.js/Express, Sequelize (PostgreSQL), JWT, Heroku

- Contributed to backend development in a 4-person full stack Agile Scrum team, implementing Sequelize models, Express API routes, and JWT authentication, streamlining course registration to improve course scheduling compared to the existing CUNYFirst system.

Experience

Software Engineer Fellow | 06/2024 – 05/2025

Springboard | Remote

- Built and deployed a full-stack community platform (Himalayan Connect NYC) using Next.js, translating Figma designs into a fully functional application that enables seamless access to key community resource. **Live**
 - Implemented robust features including authentication, dashboards, search/filter, interactive Google Maps, Redis caching, and Cloudinary, validated through unit and integration tests to ensure reliability.
 - Deployed with full documentation, achieving scalable and performant delivery by leveraging speed insights, observability tools, firewall, and rate limiting for enhanced security and reliability.
 - **Technologies:** Next.js, React, TypeScript, PostgreSQL, Prisma, Tailwind, Material UI
- Developed secure RESTful API for a dog adoption platform using Node.js, Express and MongoDB with token authentication, 7 CRUD routes, MVC design, and Mocha/Chai tests, ensuring scalable data management.
- Engineered a Space Travel Planner app using React Hooks and React Router with dynamic forms, client-side routing, React Testing Library delivering a seamless and fully tested user interface.

Tech Operations Associate | 11/2021 – 10/2022**Krishnamurthy Tandon Foundation | New York, NY**

- Managed vendor relationships, coordinating deliverables, timelines, and third-party integrations, demonstrating project management, cross-functional collaboration, and process optimization.
- Streamlined digital systems and data organization, optimizing contact databases, records, and project documentation, to enhance efficiency, accuracy, and information accessibility.
- Led digital initiatives including social media management, ad campaigns, website updates, and technical setup for events and studio operations, showing problem-solving, adaptability, and hands-on technical skills.

Data Engineer Intern | 06/2019 – 08/2019**Real Factors (Acquired by 7Park Data) | New York, NY**

- Resolved map display errors by correctly parsing geospatial data using SQL in BigQuery, ensuring accurate visualization before key acquisition deadline.
- Integrated and transformed multi-source data using SQL utilizing joins, partitions, aggregations and GeoJSON to create reliable tables and views for business intelligence and analytics.
- Developed interactive dashboard and BI tools in Apache Superset, providing actionable insights through charts, reports, while documenting processes for future team continuity.

Machine Learning Researcher (with Prof. Jia Xu) | 09/2018 – 05/2019**Hunter College | New York, NY**

- Trained and optimized a Neural Attention Model (Codenn) using PyTorch on CUNY'S HPC, leveraging CUDA and Linux shell scripting to enhance performance and efficiency.
- Researched and analyzed state-of-the-art NLP models for machine translation presenting findings to peers and professors to guide experimental design and implementation strategies.
- Documented insights and experimental outcomes, improving understanding of deep learning frameworks and contributing to group knowledge on HPC-based model optimization.

Education

Bachelor of Arts in Computer Science, Hunter College, 2020

CUNY 2X TTP Residency, Hunter College